Microsoft Purview for Manufacturing

Implementation Solution Guide

Version 1.0.0 · 2026-07-07

Contents

Executive Summary — Microsoft Purview for Manufacturing 5

What you get . . . 5

What this delivers . . . 5

The standard deployment model . . . 5

Scope boundary . . . 5

Outcome . . . 6

1. Solution Overview — Microsoft Purview for Manufacturing 6

What this delivers . . . 6

Why Manufacturing is different . . . 6

How it works in Purview . . . 6

What you can control . . . 7

Benefits . . . 7

Limitations (be honest with buyers) . . . 7

The standard deployment model . . . 8

2. Requirements Guide — Microsoft Purview for Manufacturing 8

2.1 Microsoft 365 + Purview compliance licensing (IP + CUI features) . . . 8

2.2 Least-privilege Purview roles . . . 8

2.3 Manufacturing data classification decided (IP + CUI + export control) . . . 8

2.4 Custom classification building blocks (part/drawing numbers, CUI) . . . 9

2.5 Pilot scope (users and, for endpoint DLP, devices) . . . 9

2.6 Supplier / external-sharing model agreed . . . 9

2.7 Audit logging enabled . . . 9

2.8 Records schedule & CMMC/NIST 800-171 requirements captured . . . 10

2.9 Change control for enforce-mode changes . . . 10

Checklist . . . 10

Global Requirements — Microsoft Purview Solutions 10

1. Microsoft 365 tenant with the required Purview/compliance licensing . . . 11

2. Purview roles (least-privilege) and portal access . . . 11

3. Security & Compliance PowerShell + Graph toolchain . . . 11

4. Data sources & scope in scope . . . 12

5. Compliance, records, privacy & legal stakeholders . . . 12

6. A pilot scope and change control . . . 12

3. Implementation Guide — Microsoft Purview for Manufacturing 13

Step 1 — Design the Manufacturing taxonomy (Assess/Design) . . . 13

Step 2 — Create the sensitivity labels . . . 14

Step 3 — Publish the labels with a label policy . . . 14
Step 4 — Build the custom sensitive-information type (IP detection) . . . 15
Step 5 — Configure auto-labeling in SIMULATION . . . 15
Step 6 — Build the DLP policy in AUDIT mode . . . 15
Step 7 — Pilot and tune (the DLP audit gate) . . . 16
Step 8 — Switch DLP to ENFORCE (after the gate) . . . 16
Step 9 — Create retention labels (quality / contract / CUI records) . . . 17
Step 10 — Create a retention policy + records file plan . . . 17
Step 11 — Track CMMC / NIST 800-171, then validate and expand . . . 17
5. Testing Guide — Microsoft Purview for Manufacturing 18
Test 1 — Labels published and visible . . . 18
Test 2 — IP label protection travels externally . . . 18
Test 3 — CUI / export-control label restricts . . . 18
Test 4 — Custom SIT detects a part/drawing number . . . 18
Test 5 — Auto-labeling simulation matches . . . 18
Test 6 — DLP is auditing (before the gate) . . . 18
Test 7 — DLP enforces (after the gate) . . . 19
Test 8 — Retention label applied and honored . . . 19
Test 9 — Retention policy scoped . . . 19
Test 10 — CMMC / 800-171 evidence + coverage report . . . 19
Pass/fail gate . . . 19
6. Troubleshooting Guide — Microsoft Purview for Manufacturing 20
A label or feature is missing from the portal . . . 20
New labels don't show in Office / Outlook for pilot users . . . 20
The custom SIT matches too much or too little . . . 20
Auto-labeling simulation shows nothing (or wrong labels) . . . 20
DLP policy tips don't appear / no audit events . . . 21
DLP blocks legitimate supplier / engineering work . . . 21
Label encryption breaks a legacy CAD / engineering app . . . 21
Endpoint DLP rules don't fire (USB / print / upload) . . . 21
Retention isn't applying (or deleted too much) . . . 21
CMMC / NIST 800-171 assessment template not available . . . 22
PowerShell can't connect or a cmdlet is missing . . . 22
7. Best Practices — Microsoft Purview for Manufacturing 23
8. Deliverables Summary — Microsoft Purview for Manufacturing 24
Documentation . . . 24

PowerShell scripts 24
What the buyer receives 25
Outcome 25
Appendix A — PowerShell Scripts 26

Executive Summary — Microsoft Purview for Manufacturing

White-label · Purview Solutions

Manufacturers carry two data obligations at once: the **intellectual property** that *is* the business — CAD/CAM designs, process specs, formulas, bills of material, firmware — and, for the defense supply chain, **Controlled Unclassified Information (CUI)** that must be protected to **CMMC / NIST SP 800-171** (including export-controlled **ITAR / EAR** technical data). Left to generic defaults, that data leaks through supplier email, USB drives, and browser uploads, and a CMMC assessment finds no evidence the controls exist. Microsoft Purview fixes this — but rolled out carelessly, labels confuse engineers, Data Loss Prevention (DLP) floods the help desk with false positives, and retention deletes too much or nothing at all.

This is a complete, deploy-it-in-order package that stands up a **Manufacturing-tailored Purview program**: an IP/CUI sensitivity-label taxonomy whose protection **travels with the file** to suppliers and OEMs, a **custom classifier for your part/drawing numbers**, DLP that starts in audit and is tuned before it enforces (with export-control hard-blocks and endpoint coverage), defensible quality/contract/CUI retention, and **CMMC / NIST 800-171 evidence** tracked in Compliance Manager. It is written for an administrator with no prior Purview experience.

What you get

- A plain-language Solution Overview of the IP + CUI program and how sensitivity labels, a custom SIT, auto-labeling, DLP, and retention/records work together for Manufacturing.
- A Requirements Guide with verify / implement-if-missing for every Manufacturing prerequisite.
- An Implementation Guide of precise, ordered steps in the Microsoft Purview portal.
- Five PowerShell scripts for configuration discovery, coverage validation, a classification and DLP event report, safe remediation of gaps, and a pilot-scope group builder.
- Testing and Troubleshooting guides for a confident rollout.

What this delivers

A named, reusable set of objects — an IP/CUI label taxonomy, a `SIT - MFG Part/Drawing Number` classifier, auto-labeling (simulated first), a `DLP Policy - Manufacturing IP and CUI` policy, and quality/contract retention — deployed on the assess-design-pilot model with a mandatory **DLP audit gate** before any policy switches to enforce, and a documented rollback.

The standard deployment model

Assess → Design → Configure → Pilot (simulate/test) → Validate → Expand (enforce).

Scope boundary

The solution provides the **technical configuration and evidence**. **CMMC / CUI applicability, the export-control (ITAR/EAR) determination, retention schedules, and records file plans are legal and compliance determinations** deferred to the organization's compliance, records, privacy, legal, and export-control functions.

Outcome

Manufacturing IP and CUI classified, protected, retained, and evidenced with Microsoft Purview — protection that travels to suppliers, DLP tuned in audit before it enforces, and CMMC / NIST 800-171 readiness — proven on a pilot before you scale.

1. Solution Overview — Microsoft Purview for Manufacturing

What this delivers

This solution stands up a **Manufacturing-tailored Microsoft Purview program** — sensitivity labels, Data Loss Prevention (DLP), retention/records, and compliance posture — aligned to the two data obligations that define the sector: **protecting intellectual property (IP)** (CAD/CAM designs, process specs, formulas, bills of material, firmware/source) and **safeguarding Controlled Unclassified Information (CUI)** for defense-supply-chain work under **CMMC / NIST SP 800-171** (including export-controlled technical data — **ITAR / EAR**). It is delivered end to end in Microsoft Purview: designed, configured, **piloted in simulation/test**, validated against pass/fail gates, then expanded to enforcement.

Why Manufacturing is different

- **IP is the crown jewel.** Designs, tooling, process know-how, and firmware are the business — a leaked drawing or BOM is a competitive and contractual loss, not just a compliance ticket.
- **CUI / CMMC obligations.** Manufacturers in the defense industrial base (DIB) must protect CUI to **NIST SP 800-171** and evidence it for **CMMC** assessment; several 800-171 controls map directly to Purview (media protection, access control, audit, and — with Microsoft's guidance — CUI labelling and DLP).
- **Export control.** Technical data under **ITAR / EAR** must not reach unauthorized (often foreign) persons — a DLP and labelling concern with legal weight.
- **Heavy external sharing.** OEMs, suppliers, and contract manufacturers exchange drawings and specs constantly; protection must travel with the file, not stop at the tenant boundary.

How it works in Purview

1. A small, Manufacturing-specific **sensitivity-label taxonomy** (Public → Internal → Confidential-IP → CUI, with an export-controlled variant) is published so labels appear in Office, Outlook, and the web and **travel with the file** (encryption + visual marking).
1. **Auto-labeling and custom classifiers** help identify IP and CUI at scale (a custom sensitive-information type for drawing/part numbers; the CUI trainable classifier where licensed), run first in **simulation**.
1. **DLP policies** detect CUI, export-controlled data, and IP markers across Exchange, SharePoint,

OneDrive, Teams, and managed endpoints — deployed **audit-first**, tuned, then enforced.

1. **Retention & records** keep quality, contract, and CUI records for the required period and dispose of them defensibly; **records management** makes regulatory records immutable with disposition review.

1. **Compliance Manager** tracks the **CMMC / NIST 800-171** improvement actions so the program is evidenced, not just configured.

What you can control

• The label taxonomy, label order/priority, and the protection each label applies (encryption, marking, sharing restrictions) — tuned for IP and CUI.

• Where labels publish and whether **auto-labeling / trainable classifiers** apply them by content.

• DLP scope (Exchange, SharePoint, OneDrive, Teams, endpoint devices), the sensitive-information types / labels a policy matches (CUI, ITAR/EAR, IP part-number patterns), and the action (audit → notify → block).

• Retention duration, trigger, and end-of-life action (delete, retain, or declare a record), plus disposition review for records.

Benefits

• **IP that travels protected** — a label encrypts and marks a drawing or spec so it stays protected when shared to a supplier or OEM.

• **CUI safeguarded and evidenced** — DLP + labels + audit map to NIST 800-171 / CMMC and produce the evidence an assessor expects.

• **Fewer leaks, fewer false positives** — DLP tuned in audit before it blocks real risk without blocking legitimate manufacturing work.

• **Defensible retention** — quality and contract records kept and disposed on a defined schedule.

Limitations (be honest with buyers)

• Auto-labeling at rest, endpoint DLP, trainable classifiers, and advanced records require the higher **Microsoft 365 E5 / E5 Compliance** plans; the Requirements Guide names the licensing — **verify your licensing/tenant**.

• **CMMC / CUI applicability, the export-control determination (ITAR/EAR), retention schedules, and records file plans are legal/compliance determinations **— this solution provides the technical method and evidence,** not** the determination; those are deferred to your compliance, records, privacy, legal, and export-control functions.

• Purview protects data **in Microsoft 365**; OT/shop-floor systems, PLM/PDM, and CAD file shares

outside M365 are out of scope unless their data lands in M365 workloads.

• Encryption applied by a label can affect legacy CAD/engineering apps that do not understand the protection — **pilot before broad enforcement**.

The standard deployment model

> ***Assess*** → ***Design*** → ***Configure*** → ***Pilot (simulate/test)*** → ***Validate*** → ***Expand (enforce)***, *with a* ***DLP audit gate*** *before any policy enforces, simulation for auto-labeling, and a documented rollback.*

2. Requirements Guide — Microsoft Purview for Manufacturing

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-purview.md` (licensing, least-privilege roles, S&C PowerShell/Graph toolchain, in-scope data, compliance/records/legal stakeholders, and a pilot scope with change control) — this guide adds the **Manufacturing-specific** prerequisites and does not repeat the six.

2.1 Microsoft 365 + Purview compliance licensing (IP + CUI features)

Description: Licensing that includes the capabilities this program uses. Sensitivity labels and baseline DLP ship in **Microsoft 365 E3 / Business Premium**; **auto-labeling at rest, endpoint DLP, trainable classifiers, advanced records management, and Compliance Manager premium assessments** require **Microsoft 365 E5** (or the **E5 Compliance / Information Protection & Governance** add-on). DIB tenants often run in **GCC / GCC High** — confirm the CUI-appropriate cloud. **Why:** The IP/CUI features here are plan-gated and cloud-gated; they will not appear without the right plan (and, for CUI, often the government cloud). **Verify:** M365 admin center → **Billing** → **Licenses**; confirm the feature is visible in the Purview portal; run `PURVIEW-MFG-Discovery.ps1`. **Verify your licensing/tenant** — availability evolves. **Implement:** Assign the required plan (E3/Business Premium for labels + basic DLP; **E5 or the E5 Compliance add-on** for auto-labeling at rest, endpoint DLP, trainable classifiers, advanced records, Compliance Manager). Confirm the correct sovereign cloud with your compliance/legal team for CUI.

2.2 Least-privilege Purview roles

Description: The operator holds the least-privilege roles to create labels, DLP, retention, and to read compliance posture — for example **Compliance Administrator** (or **Information Protection Admin** plus **Data Loss Prevention** roles) and, for records/disposition, **Records Management**; **Compliance Manager** read/assessment roles for the CMMC template. **Why:** Purview uses fine-grained role groups; Global Administrator alone is not the least-privilege choice. **Verify:** `https://purview.microsoft.com` → **Settings** → **Roles and scopes** → **Role groups**; confirm membership. **Implement:** Assign the least-privilege role group under change control; reserve Global Administrator for tenant-wide changes.

2.3 Manufacturing data classification decided (IP + CUI + export control)

Description: A one-page classification decision from the business: what is **IP** (designs, CAD/CAM, process specs, formulas, BOMs, firmware/source), what is **CUI** (per the contract / DFARS 252.204-7012),

and what is **export-controlled** (**ITAR / EAR**) — feeding the label taxonomy this solution publishes. **Why:** Labels, DLP conditions, and retention are business/legal decisions, not IT defaults; the CUI and export-control determinations are made by compliance/legal/export-control, not by IT. **Verify:** A named data owner exists per category and a marked CUI/export-control determination is recorded (or explicitly deferred to the responsible function). **Implement:** Convene data owners + compliance/legal/export-control to agree the categories and markings **before** authoring policies (Implementation Step 1). **Determinations are deferred** to those functions.

2.4 Custom classification building blocks (part/drawing numbers, CUI)

Description: The identifiers that flag Manufacturing IP — e.g. a **drawing/part-number pattern** — captured so a **custom sensitive-information type (SIT)** can detect them, plus a decision on using the **CUI trainable classifier / CUI SITs** (availability varies by plan/cloud — **verify your tenant**). **Why:** Generic financial/PII detectors miss engineering IP; a custom SIT keyed to your part-number format materially improves DLP and auto-labeling accuracy. **Verify:** A sample of real part/drawing numbers and their format (regex-able) exists; confirm CUI classifier availability in the Purview portal. **Implement:** Provide sample identifiers so `SIT - MFG Part/Drawing Number` can be built (Implementation Step 4); confirm CUI SIT/classifier availability before relying on it.

2.5 Pilot scope (users and, for endpoint DLP, devices)

Description: A Microsoft Entra security group of pilot users — engineering + a CUI/contracts team — named `Purview - Manufacturing Pilot`, and, for endpoint DLP, **engineering workstations onboarded to Microsoft Purview** (the same onboarding used by Defender for Endpoint). **Why:** Every policy is piloted before rollout; DLP must run in audit on a pilot first, and endpoint rules (copy to USB, print a drawing, upload to a browser) require onboarded devices. **Verify:** Entra → **Groups** shows the group; Purview → **Settings** → **Device onboarding** → **Devices** shows pilot devices (endpoint DLP only); run `PURVIEW-MFG-Discovery.ps1`. **Implement:** Build the group with `PURVIEW-MFG-Set-ScopeTargeting.ps1`; onboard pilot engineering devices via the Purview onboarding package (or Intune).

2.6 Supplier / external-sharing model agreed

Description: How drawings and specs are shared externally (OEMs, suppliers, contract manufacturers) — the collaboration boundaries the labels/DLP must respect (e.g. **encrypt-and-restrict to named partner domains**, block export-controlled data to unauthorized recipients). **Why:** Manufacturing lives on external collaboration; protection that blocks legitimate supplier sharing will be bypassed, while unrestricted sharing leaks IP/CUI. **Verify:** A list of trusted partner domains and the allowed sharing pattern per label exists. **Implement:** Capture partner domains and the per-label sharing rule with the business before configuring label encryption and DLP recipient conditions.

2.7 Audit logging enabled

Description: Microsoft Purview **audit** (unified audit log) is on — the evidence source for DLP tuning, CUI access, and CMMC audit-control evidence. **Why:** DLP tuning, coverage reporting, and 800-171/CMMC audit evidence read audit/activity data. **Verify:** Purview → **Audit**; searching returns results; run `PURVIEW-MFG-Report.ps1`. **Implement:** Turn on auditing in the Purview **Audit** solution and allow time for events to populate.

2.8 Records schedule & CMMC/NIST 800-171 requirements captured

Description: The **retention schedule** (quality records, contract/CUI records, export records) and the **CMMC / NIST 800-171** control set the program must evidence — sourced from records management, compliance, and the contract (DFARS), not guessed. **Why:** Retention durations and records determinations are legal/records decisions; CMMC control scope drives which Purview controls you evidence in Compliance Manager. **Verify:** A records schedule source and the applicable CMMC level / 800-171 baseline are documented with named approvers. **Implement:** Obtain the schedule and control baseline from records/compliance; load the **NIST 800-171 / CMMC** assessment template in **Compliance Manager** (Implementation Step 10). **Retention and records determinations are deferred** to records/compliance/legal.

2.9 Change control for enforce-mode changes

Description: A documented change-control / rollback process for switching any DLP policy from audit to enforce, turning auto-labeling out of simulation, or activating a retention policy that deletes content. **Why:** Enforce-mode DLP can block engineering/supplier work; auto-labeling mistakes are user-visible; retention deletion is irreversible. **Verify:** A change ticket exists with an approver, a pilot scope, and a documented rollback. **Implement:** Record the change, name the pilot scope, define the rollback, and get approval before enforcing.

Checklist

- M365 + Purview licensing (E3/Business Premium for labels + basic DLP; **E5 / E5 Compliance** for auto-labeling at rest, endpoint DLP, trainable classifiers, advanced records, Compliance Manager); correct sovereign cloud for CUI — `PURVIEW-MFG-Discovery.ps1`
- Least-privilege Purview roles assigned (Compliance Admin / Information Protection + DLP; Records Management)
- Manufacturing classification decided — IP / CUI / export-controlled (determinations by compliance/legal/export-control)
- Custom SIT identifiers (part/drawing numbers) supplied; CUI classifier availability confirmed
- Pilot group `Purview - Manufacturing Pilot` created; engineering devices onboarded (endpoint DLP) — `PURVIEW-MFG-Set-ScopeTargeting.ps1`
- Supplier / external-sharing model + trusted partner domains agreed
- Purview audit logging enabled — `PURVIEW-MFG-Report.ps1`
- Records schedule + CMMC/NIST 800-171 baseline captured (records/compliance)
- Change control + rollback for enforce-mode / auto-labeling / retention deletion

Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication

compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy, and legal functions** — the solutions provide the technical configuration method and evidence, not the determination. Automation is **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with the required Purview/compliance licensing

• **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

• **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

• **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

• **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Purview roles (least-privilege) and portal access

• **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

• **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.

• **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.

• **How to implement if missing:** Assign the least-privilege role group under change control.

3. Security & Compliance PowerShell + Graph toolchain

• **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

• **Why required:** Configuration, automation, and the provided scripts run through these.

- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams, endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.
- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.
- **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with stakeholders.
- **How to implement if missing:** Define the scope and classification intent before authoring policies.

5. Compliance, records, privacy & legal stakeholders

- **Description:** Engagement with the **compliance, records-management, privacy, security, and legal** stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).
- **Why required:** Purview configuration must align to enterprise policy and legal/records requirements, not technical defaults; several capabilities have HR/legal sensitivity.
- **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.
- **How to implement if missing:** Identify the owners and capture the requirements before configuring.

6. A pilot scope and change control

- **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.
- **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate → Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.
- **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change window/approval path.
- **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant

governance functions.*

3. Implementation Guide — Microsoft Purview for Manufacturing

Follow these steps **in order**, in the **Microsoft Purview portal** (`https://purview.microsoft.com`). Portal paths and cmdlets are current at authoring; Purview evolves and features are licensing-/cloud-dependent — **verify your licensing/tenant** and treat any absent feature as a plan/cloud gap (Requirement 2.1), not a mistake in the steps.

> ***Model:*** *Design a Manufacturing taxonomy (IP + CUI) → publish labels → add auto-labeling / custom classifiers in* ***simulation*** *→ build DLP in* ***audit*** *and tune it (the* ***DLP audit gate****) before enforcing → set retention/records → track CMMC/800-171 in Compliance Manager — piloting each on* `Purview - Manufacturing Pilot` *before you expand. Pattern:* ***Assess*** *→* ***Design*** *→* ***Configure*** *→* ***Pilot (simulate/test)*** *→* ***Validate*** *→* ***Expand (enforce)****.*

The named objects created below are reused verbatim in Testing, Troubleshooting, and the scripts:

Object	Name
Sensitivity labels	`Sensitivity Label - MFG-Public`, `Sensitivity Label - MFG-Internal`, `Sensitivity Label - MFG-Confidential-IP`, `Sensitivity Label - MFG-CUI`, `Sensitivity Label - MFG-CUI-Export-Controlled`
Label policy	`Label Policy - Manufacturing Taxonomy - All Users`
Custom SIT	`SIT - MFG Part/Drawing Number`
Auto-labeling policy	`Auto-Label - CUI and Export-Controlled - Pilot`
DLP policy	`DLP Policy - Manufacturing IP and CUI - Audit` (renamed to `... - Enforce` at the gate)
Retention label	`Retention Label - Quality and Contract Records - 7 Year`
Retention policy	`Retention Policy - Manufacturing Baseline - 5 Year`
Compliance assessment	`Assessment - CMMC / NIST 800-171 - Manufacturing`

Step 1 — Design the Manufacturing taxonomy (Assess/Design)

1. With data owners **and** compliance/legal/export-control, agree the labels and the protection each applies. A Manufacturing-tailored starting set:

• `Sensitivity Label - MFG-Public` — cleared for public release; no protection.

• `Sensitivity Label - MFG-Internal` — internal business data; no encryption.

• `Sensitivity Label - MFG-Confidential-IP` — designs/specs/BOMs/firmware; **encrypt**, restrict to

the organization **and named partner domains** (Requirement 2.6).

• `Sensitivity Label - MFG-CUI` — Controlled Unclassified Information; **encrypt**, restrict to

authorized users; **CUI header/footer marking**.

• `Sensitivity Label - MFG-CUI-Export-Controlled` — ITAR/EAR technical data; **encrypt**, restrict

to authorized **U.S. persons / named recipients only**; export-control marking.

1. Record each label's meaning, protection, marking, and who applies it. **The CUI and export-control

determinations are made by compliance/legal/export-control** — capture them, don't invent them.

Step 2 — Create the sensitivity labels

1. Purview portal → **Information Protection** → **Sensitivity labels** → **Create a label**.

2. **Name and tooltip:** enter the label name (e.g. `Sensitivity Label - MFG-CUI`) and a user-facing

tooltip. **Next**.

1. **Scope:** select **Items** (files and emails). **Next**.

2. **Encryption:** choose **Configure encryption** for `MFG-Confidential-IP`, `MFG-CUI`, and

`MFG-CUI-Export-Controlled`; set permissions — for `MFG-Confidential-IP` allow the org **and** the named partner domains; for the CUI labels restrict to authorized users/recipients only. Leave `MFG-Public` / `MFG-Internal` unencrypted. **Next**.

1. **Content marking:** add a header/footer/watermark where required — e.g. `CUI` for `MFG-CUI` and an

export-control banner for `MFG-CUI-Export-Controlled`. **Next**.

1. **Auto-labeling for items:** leave off here (configured centrally in Step 5). **Next → Create the

label**.

1. Repeat for each label, creating them in sensitivity order

(`MFG-Public` → `MFG-Internal` → `MFG-Confidential-IP` → `MFG-CUI` → `MFG-CUI-Export-Controlled`). Label order sets priority.

Step 3 — Publish the labels with a label policy

1. Purview portal → **Information Protection** → **Sensitivity labels** → **Label policies** → **Publish labels**.

2. **Choose labels to publish:** add all five taxonomy labels. **Next**.

3. **Assign to users and groups:** add `Purview - Manufacturing Pilot` for the pilot. **Next**.

4. **Policy settings:** require a **justification for downgrading** a label and (optionally) **require a label for new items — important for IP/CUI hygiene.** Next**.

1. **Default label / mandatory labeling:** set a default (for example `MFG-Internal`) if the org wants one. **Next**.

1. **Name:** `Label Policy - Manufacturing Taxonomy - All Users`. **Next** → **Submit**.

Step 4 — Build the custom sensitive-information type (IP detection)

1. Purview portal → **Data classification → Classifiers → Sensitive info types → Create sensitive info type**.

1. **Name:** `SIT - MFG Part/Drawing Number`. **Next**.

2. **Pattern:** add a **primary element** as a **regular expression** matching your part/drawing-number format (from Requirement 2.4), optionally with **supporting keywords** (e.g. `drawing`, `rev`, `assembly`, `BOM`). Set a **confidence level** and **character proximity**. **Next** → **Create**.

1. Test it with **Test** using sample content before relying on it. (For CUI, also confirm whether the built-in **CUI trainable classifier / CUI SITs** are available in your plan/cloud — **verify your tenant** — and prefer them where present.)

Step 5 — Configure auto-labeling in SIMULATION

1. Purview portal → **Information Protection** → **Auto-labeling** → **Create auto-labeling policy**.

2. **Choose the information to label:** pick **Custom** (to use `SIT - MFG Part/Drawing Number` and/or the CUI classifier). **Next**.

1. **Name:** `Auto-Label - CUI and Export-Controlled - Pilot`. **Next**.

2. **Locations:** select **SharePoint sites**, **OneDrive accounts**, and **Exchange email** as needed; scope to pilot locations. **Next**.

1. **Rules:** condition = `SIT - MFG Part/Drawing Number` (and/or the CUI classifier); set the label to apply — `Sensitivity Label - MFG-Confidential-IP` for IP, `Sensitivity Label - MFG-CUI` for CUI matches. **Next**.

1. **Test the policy first:** choose **Run policy in simulation mode** — do **not** turn it on. **Next → Create**. Review simulation matches (Step 7 / Testing) before turning it on.

Step 6 — Build the DLP policy in AUDIT mode

1. Purview portal → **Data Loss Prevention** → **Policies** → **Create policy**.

2. **Category / template:** choose **Custom** (Manufacturing IP/CUI is not a stock template). **Next**.

3. **Name:** `DLP Policy - Manufacturing IP and CUI - Audit`. **Next**.

4. **Locations:** enable **Exchange email**, **SharePoint sites**, **OneDrive accounts**, **Teams chat and channel messages**, and** Devices **(endpoint DLP) as licensed. Scope to** `Purview - Manufacturing Pilot` **where the location supports user scoping.** Next**.

1. **Rules:** add rules that match:

• the label `Sensitivity Label - MFG-CUI` **or** `Sensitivity Label - MFG-Confidential-IP`,

• `SIT - MFG Part/Drawing Number`, and

• export-control indicators (the **ITAR / EAR** sensitive-information types where available — **verify your tenant**), with a** recipient condition **blocking send to domains outside the trusted partner list (Requirement 2.6). Set actions to** audit only **— enable** policy tips **for visibility but do** not **block. For endpoint rules, audit** copy to USB / removable media, print, and upload to unallowed browsers**.** Next**.

1. **Policy mode:** select **Run the policy in test mode** (policy tips on). **Next** → **Submit**. The policy now audits without blocking.

Step 7 — Pilot and tune (the DLP audit gate)

1. Let the audit-mode policy run against `Purview - Manufacturing Pilot` for the agreed window.
2. Review matches in Purview → **Data Loss Prevention** → **Alerts** and **Activity explorer**; run `PURVIEW-MFG-Report.ps1` for a classification and DLP-event summary. Review the auto-labeling **simulation** results too.

1. Tune to remove false positives — adjust the SIT confidence/proximity, instance counts, and exceptions (e.g. legitimate supplier domains).

1. **Gate:** only when the audit shows the policy matches real IP/CUI/export-controlled data and not routine engineering work do you proceed to enforce (Step 8). If it is still noisy, keep tuning — **do not enforce**.

Step 8 — Switch DLP to ENFORCE (after the gate)

1. Purview portal → **Data Loss Prevention** → **Policies** → open `DLP Policy - Manufacturing IP and CUI - Audit` → **Edit**.
1. Rename it to `DLP Policy - Manufacturing IP and CUI - Enforce` so the stage is visible.
2. **Rules:** change the action to **Block** (or **Block with override** for internal recipients) as the business requires; keep the export-control rule as **hard block**. Keep policy tips on. **Next**.

1. **Policy mode:** select **Turn the policy on**. **Next** → **Submit**. Enforcement applies to the pilot scope first; widen in Step 11.

1. **Rollback (if needed):** set the policy mode back to **Test mode** (audit) or turn the policy off.

Step 9 — Create retention labels (quality / contract / CUI records)

1. Purview portal → **Data Lifecycle Management** → **Retention labels** → **Create a label**.
2. **Name:** `Retention Label - Quality and Contract Records - 7 Year`. **Next**.
3. **Retention settings: Retain items for** the period from your records schedule (Requirement 2.8;

7 years shown as a placeholder — **the schedule is a records/compliance determination**), start of period based on when items were created (or labeled). End action: **Delete**, or **Mark as a record** (records management) where the item must be immutable (e.g. CUI/contract records). **Next** → **Create**.

Step 10 — Create a retention policy + records file plan

1. Purview portal → **Data Lifecycle Management** → **Retention policies** → **New retention policy**.
2. **Name:** `Retention Policy - Manufacturing Baseline - 5 Year`. **Next**.
3. **Locations:** enable **Exchange mailboxes**, **SharePoint sites**, **OneDrive accounts**, and

Teams as required. Scope to `Purview - Manufacturing Pilot` for the pilot. **Next**.

1. **Retention settings:** retain for the baseline period from the schedule, then delete or retain-only.

Next → **Submit**.

1. For regulatory/CUI records, configure **Records Management** → **File plan** to publish record labels and

enable **disposition review** so deletions are reviewed, not automatic. **File-plan content and schedules are records/compliance determinations.**

Step 11 — Track CMMC / NIST 800-171, then validate and expand

1. Purview portal → **Compliance Manager** → **Assessments** → **Add assessment**; select the **CMMC** /

NIST SP 800-171 template (availability varies by plan/cloud — **verify your tenant**) and name it `Assessment - CMMC / NIST 800-171 - Manufacturing`. Work the **improvement actions** that map to the labels/DLP/retention/audit you configured; capture the evidence.

1. Complete the Testing Guide against `Purview - Manufacturing Pilot`; run `PURVIEW-MFG-Validation.ps1`

(labels published, DLP present, retention present) and `PURVIEW-MFG-Report.ps1` (coverage).

1. **Expand:** republish `Label Policy - Manufacturing Taxonomy - All Users`, the DLP **Enforce**

policy, and the retention policy from the pilot to broader groups (or **All users** / **All locations**) once the pilot passes. Turn auto-labeling **on** (out of simulation) only after its simulation is clean.

1. Re-validate each expansion with `PURVIEW-MFG-Validation.ps1` and `PURVIEW-MFG-Report.ps1`.

Order of operations

`Taxonomy (IP+CUI) → labels → label policy → custom SIT → auto-labeling (simulation) → DLP (audit) → audit gate → DLP (enforce) → retention labels → retention policy/records → CMMC/800-171 in Compliance Manager → validate → expand`

5. Testing Guide — Microsoft Purview for Manufacturing

Run after publishing to `Purview - Manufacturing Pilot` (Implementation Step 3 onward), before the DLP audit gate (Step 7), and again after each expansion.

Test 1 — Labels published and visible

Do: In Outlook / Word (web or desktop) as a pilot user, open the **Sensitivity** button. Or run `PURVIEW-MFG-Validation.ps1`. **Expected:** All five taxonomy labels appear (`Sensitivity Label - MFG-Public` through `Sensitivity Label - MFG-CUI-Export-Controlled`) via `Label Policy - Manufacturing Taxonomy - All Users`.

Test 2 — IP label protection travels externally

Do: Apply `Sensitivity Label - MFG-Confidential-IP` to a test drawing and share it to (a) a listed trusted partner domain and (b) a non-listed external address. **Expected:** The file is encrypted/marked; the trusted partner can open per the label's permissions; the non-listed external recipient is denied.

Test 3 — CUI / export-control label restricts

Do: Apply `Sensitivity Label - MFG-CUI` (and separately `MFG-CUI-Export-Controlled`) to a test item and attempt to open it as an out-of-scope/unauthorized user. **Expected:** The item is encrypted and shows the CUI / export-control marking; the unauthorized user is denied per the label's permissions.

Test 4 — Custom SIT detects a part/drawing number

Do: Purview → **Data classification** → **Sensitive info types** → open `SIT - MFG Part/Drawing Number` → **Test**, using sample content containing a real-format part number and some decoy text. **Expected:** The SIT matches the part/drawing number at the configured confidence and does **not** match the decoys.

Test 5 — Auto-labeling simulation matches

Do: Purview → **Information Protection** → **Auto-labeling** → open `Auto-Label - CUI and Export-Controlled - Pilot` → **Simulation**. **Expected:** The simulation lists items containing part/drawing numbers or CUI markers and proposes the correct label — and does not flag routine, non-sensitive content.

Test 6 — DLP is auditing (before the gate)

Do: As a pilot user, email a test message containing a labelled IP/CUI file (or a part number) to an external address; check Purview → **Data Loss Prevention** → **Alerts** and **Activity explorer**. Or run

`PURVIEW-MFG-Report.ps1`. **Expected:** `DLP Policy - Manufacturing IP and CUI - Audit` records the match (policy tip shown, no block) — the audit gate has data to review.

Test 7 — DLP enforces (after the gate)

Do: After Step 8, repeat Test 6 against `DLP Policy - Manufacturing IP and CUI - Enforce`; also try to send an export-controlled item to an unauthorized recipient, and (endpoint) copy a labelled drawing to a USB drive. **Expected:** The action now blocks (or blocks-with-override for internal); the export-control send is hard-blocked; the USB copy is blocked/audited per the endpoint rule; each event is logged.

Test 8 — Retention label applied and honored

Do: Apply `Retention Label - Quality and Contract Records - 7 Year` to a test item and attempt to delete it. **Expected:** The item is retained for the period (or declared a record and blocked from deletion) per the label.

Test 9 — Retention policy scoped

Do: Confirm `Retention Policy - Manufacturing Baseline - 5 Year` is applied to the pilot locations (Purview → **Data Lifecycle Management** → **Retention policies** → open the policy). Or run `PURVIEW-MFG-Validation.ps1`. **Expected:** The policy shows **On** and scoped to the pilot mailboxes/sites/OneDrive/Teams.

Test 10 — CMMC / 800-171 evidence + coverage report

Do: In **Compliance Manager**, open `Assessment - CMMC / NIST 800-171 - Manufacturing` and confirm the configured controls show progress/evidence. Run `PURVIEW-MFG-Report.ps1`. **Expected:** The mapped improvement actions reflect the labels/DLP/retention/audit you deployed; the report summarizes classification and DLP events across the enabled services so coverage and gaps are visible.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–6, 8–9 pass on `Purview - Manufacturing Pilot`; the DLP audit shows real IP/CUI matches and acceptable noise
Audit gate	DLP audit reviewed and tuned; auto-labeling simulation clean; only then is Test 7 (enforce) run
Expand pass	Tests 1, 6/7, 9, 10 pass after widening scope beyond the pilot

6. Troubleshooting Guide — Microsoft Purview for Manufacturing

Common field issues for the Manufacturing IP + CUI program and how to resolve them. Objects are named verbatim from the Implementation Guide. Most problems are **licensing/cloud gaps**, **replication delay**, or **audit-not-yet-tuned** — not step mistakes. When a feature is simply absent, treat it as a plan/cloud gap (Requirement 2.1) and **verify your licensing/tenant** before assuming a fault.

A label or feature is missing from the portal

• **Symptom:** Auto-labeling at rest, endpoint DLP, trainable classifiers, or advanced records don't

appear.

• **Likely cause:** These need **Microsoft 365 E5 / E5 Compliance**; DIB tenants often need **GCC /

GCC High** for CUI. Basic labels + DLP ship in E3/Business Premium.

• **Resolve:** Run `PURVIEW-MFG-Discovery.ps1` to see what the tenant exposes; confirm the plan in the

M365 admin center; assign the required add-on. **Verify your licensing/tenant** — availability evolves and is cloud-dependent.

New labels don't show in Office / Outlook for pilot users

• **Symptom:** `Sensitivity Label - MFG-*` labels aren't visible after publishing.

• **Likely cause: Replication delay** (labels/policies can take hours to appear), the user isn't in

`Purview - Manufacturing Pilot`, or the client hasn't refreshed.

• **Resolve:** Confirm publication with `PURVIEW-MFG-Validation.ps1`; verify the user is in the pilot

group (`PURVIEW-MFG-Set-ScopeTargeting.ps1`); have them restart the Office app / re-sign-in; allow time to propagate before re-publishing.

The custom SIT matches too much or too little

• **Symptom:** `SIT - MFG Part/Drawing Number` fires on ordinary text, or misses real part numbers.

• **Likely cause:** Regex too broad/narrow, wrong **confidence level**, or too-wide **character

proximity**; missing supporting keywords.

• **Resolve:** Open the SIT → **Test** with real and decoy samples; tighten the regex, raise the

confidence, add supporting keywords (`drawing`, `rev`, `assembly`, `BOM`), and reduce proximity. Where the plan provides the built-in **CUI classifier/SITs**, prefer them for CUI.

Auto-labeling simulation shows nothing (or wrong labels)

• **Symptom:** `Auto-Label - CUI and Export-Controlled - Pilot` simulation is empty or mislabels.

• **Likely cause:** Simulation needs time to crawl; the rule condition or target label is wrong; the

in-scope locations hold no matching content yet.

• **Resolve:** Allow the simulation to complete; confirm the condition uses `SIT - MFG Part/Drawing

Number` (and/or the CUI classifier) and applies the correct label; seed a pilot site with sample IP/ CUI content and re-run. Never turn the policy on until the simulation is clean.

DLP policy tips don't appear / no audit events

• **Symptom:** No policy tips or Activity-explorer events from `DLP Policy - Manufacturing IP and CUI - Audit`.

• **Likely cause:** Policy still deploying; **audit logging off** (Requirement 2.7); the location isn't in scope; endpoint devices not onboarded.

• **Resolve:** Confirm auditing is on and returning results; run `PURVIEW-MFG-Report.ps1` for the event summary; verify the policy's locations and pilot scoping; for endpoint rules, confirm devices show under **Settings** → **Device onboarding**.

DLP blocks legitimate supplier / engineering work

• **Symptom:** After enforce, real drawings/specs to trusted partners are blocked.

• **Likely cause:** The policy was enforced before the **audit gate** was clean, or the trusted partner domains (Requirement 2.6) aren't in the recipient exceptions.

• **Resolve: Rollback** — set `DLP Policy - Manufacturing IP and CUI - Enforce` back to **Test mode** (audit); add the trusted partner domains as exceptions; keep export-control as a hard block; re-tune in audit, then re-enforce only after the gate passes.

Label encryption breaks a legacy CAD / engineering app

• **Symptom:** An encrypted `MFG-Confidential-IP` / `MFG-CUI` file won't open in a legacy app.

• **Likely cause:** The application doesn't understand Microsoft Purview encryption/rights.

• **Resolve:** Pilot encryption on a small scope first; for apps that can't consume protection, adjust the label's encryption/permissions or exclude those file paths from auto-labeling; validate with data owners before broad enforcement.

Endpoint DLP rules don't fire (USB / print / upload)

• **Symptom:** Copy-to-USB, print, or browser-upload of a labelled drawing isn't audited/blocked.

• **Likely cause:** Engineering workstations aren't **onboarded** to Purview, or **Devices** isn't a location on the policy.

• **Resolve:** Onboard devices (the Defender for Endpoint / Intune onboarding), confirm they appear under device onboarding, and ensure the DLP policy includes **Devices** with the endpoint actions.

Retention isn't applying (or deleted too much)

• **Symptom:** `Retention Label - Quality and Contract Records - 7 Year` or `Retention Policy -

Manufacturing Baseline - 5 Year` doesn't behave as expected.

• **Likely cause:** Policy still propagating; wrong locations; duration/end-action from a **guess**

rather than the records schedule.

• **Resolve:** Confirm the policy is **On** and scoped (`PURVIEW-MFG-Validation.ps1`); use **disposition

review **so nothing auto-deletes unreviewed.** Retention durations and records determinations are records/compliance/legal calls** — confirm the schedule before activating deletion.

CMMC / NIST 800-171 assessment template not available

• **Symptom:** `Assessment - CMMC / NIST 800-171 - Manufacturing` can't be created.

• **Likely cause:** The premium **Compliance Manager** template needs E5 / E5 Compliance and varies by

cloud.

• **Resolve:** Confirm the plan; **verify your tenant**. Where the template is absent, evidence the same

controls against the available baseline and record the gap — **the CMMC applicability is a compliance determination**, deferred.

PowerShell can't connect or a cmdlet is missing

• **Symptom:** `Connect-IPPSSession` fails, or `Get-Label` / `Get-DlpCompliancePolicy` is unavailable.

• **Likely cause:** ExchangeOnlineManagement module missing/old, or MFA/modern-auth blocked.

• **Resolve:** `Install-Module ExchangeOnlineManagement -Scope CurrentUser`; sign in with an MFA-capable

Compliance Administrator; retry. A cmdlet that stays missing after a good connection is a plan/cloud gap — **verify your tenant.**

Escalation & rollback

Situation	Action
DLP enforce disrupts work	Revert to **Test mode** (audit); fix exceptions; re-run the audit gate
Auto-labeling mislabels	Turn the policy back to **simulation**; fix the rule; re-simulate
Retention deleting wrongly	Disable the retention policy; enable **disposition review**; confirm the schedule with records/legal
Feature genuinely absent	Treat as licensing/cloud gap (Req 2.1) — **verify your licensing/tenant**, not a step error

Situation	Action
Records / CUI / export-control question	**Determination deferred** to compliance/records/legal/export-control — capture, don't decide

7. Best Practices — Microsoft Purview for Manufacturing

• **Classify IP and CUI separately — and let the business/legal decide the lines.** Keep a small taxonomy (`MFG-Public` → `MFG-Internal` → `MFG-Confidential-IP` → `MFG-CUI` → `MFG-CUI-Export-Controlled`) people can reason about. The **CUI and export-control (ITAR/EAR) determinations are legal/compliance calls** — capture them, never invent them.

• **Protect what travels.** Manufacturing runs on external sharing; use label **encryption scoped to named partner domains** for `MFG-Confidential-IP` so a shared drawing stays protected, and restrict the CUI labels to authorized recipients only. Protection that travels beats a network boundary.

• **Build a custom SIT for your part/drawing numbers.** Generic financial/PII detectors miss engineering IP; a `SIT - MFG Part/Drawing Number` keyed to your real format is the single biggest accuracy win for DLP and auto-labeling. Prefer the built-in **CUI classifier/SITs** for CUI where your plan/cloud provides them.

• **Simulate auto-labeling.** Always run `Auto-Label - CUI and Export-Controlled - Pilot` in simulation and review matches before turning it on — it labels IP/CUI at scale and mistakes are visible to engineers and suppliers.

• **Audit DLP before you enforce.** The single most important practice: run `DLP Policy - Manufacturing IP and CUI - Audit` in test mode, review matches, tune out false positives (legitimate supplier domains, decoy part numbers), and only then enforce. A blocked-work storm on the shop floor is the fastest way to lose trust in DLP.

• **Hard-block export-controlled data; block-with-override elsewhere.** Treat ITAR/EAR sends to unauthorized recipients as a hard block; for internal IP, block-with-override plus a policy tip keeps engineers productive while capturing the risk decision.

• **Cover the endpoint.** Onboard engineering workstations so endpoint DLP can govern **copy to USB, print, and browser upload** of labelled drawings — the common real-world IP-leak paths.

• **Name every object by purpose and stage.** Use `<Purpose> - <Scope> - <Stage>` (rename `... - Audit` to `... - Enforce` at the gate) so the portal shows exactly where each policy sits in its lifecycle.

• **Match retention to the records schedule, and use disposition review.** Set durations from the records/DFARS schedule (a records/compliance determination), not a guess; use disposition review for CUI/contract records so nothing is auto-deleted unreviewed. Deletion is irreversible.

• **Evidence CMMC / NIST 800-171 as you go.** Work the mapped improvement actions in Compliance

Manager (`Assessment - CMMC / NIST 800-171 - Manufacturing`) while you configure, so the program is assessor-ready — configuration without evidence is not compliance.

• **Verify licensing and cloud for CUI.** Auto-labeling at rest, endpoint DLP, trainable classifiers, and advanced records need E5 / E5 Compliance; DIB tenants often need **GCC / GCC High** for CUI — **verify your licensing/tenant** rather than assuming a feature is present.

• **Document the objects and the rollback.** Record the taxonomy, SIT, policy names, partner domains, scopes, and the rollback (revert DLP to audit; auto-labeling back to simulation; disable a retention policy) for repeatability and change control.

8. Deliverables Summary — Microsoft Purview for Manufacturing

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What a Manufacturing-tailored Purview program is — IP protection + CUI/CMMC — how it works, benefits, limitations
2. Requirements Guide	Every Manufacturing prerequisite with Verify / Implement-if-missing (references the Purview category Global Requirements)
Global Requirements (Purview)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered Purview portal steps — IP+CUI taxonomy, custom SIT, auto-labeling (simulation), DLP audit-then-enforce, retention/records, CMMC/800-171 in Compliance Manager
5. Testing Guide	Validation procedures with pass/fail gates and the DLP audit gate
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations for IP + CUI

PowerShell scripts

Script	What it does
`PURVIEW-MFG-Discovery.ps1`	Reports the tenant's sensitivity labels, DLP policies, retention config, and the custom SIT; flags endpoint onboarding and licensing signals

Script	What it does
`PURVIEW-MFG-Validation.ps1`	Confirms the Manufacturing labels are published, the DLP policy exists (with mode), and the retention policy exists and is on
`PURVIEW-MFG-Report.ps1`	Classification + DLP-event coverage summary across the enabled services (IP/CUI focus)
`PURVIEW-MFG-Remediation.ps1`	Safe remediation of common gaps (default `-WhatIf`); enables a published-but-off label/retention policy — never auto-enforces DLP
`PURVIEW-MFG-Set-ScopeTargeting.ps1`	Builds/updates the `Purview - Manufacturing Pilot` Entra group

What the buyer receives

- This implementation guide as a Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process following the model: **Assess → Design → Configure → Pilot (simulate/test) → Validate → Expand (enforce)**, with a DLP audit gate before enforce and a documented rollback.

Outcome

A Manufacturing-tailored Microsoft Purview program — a clear IP/CUI label taxonomy that travels with the file, a custom part-number classifier, DLP tuned in audit before it enforces (with export-control hard-blocks and endpoint coverage), defensible quality/contract/CUI retention, and CMMC / NIST 800-171 evidence in Compliance Manager — proven on a pilot before you scale. **Regulatory, records, and export-control determinations remain with the organization's compliance, records, legal, and export-control functions.**

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

PURVIEW-MFG-Discovery.ps1

Purpose: Connects to Security & Compliance PowerShell and reports the tenant's current Microsoft Purview information-protection posture for a Manufacturing IP + CUI program: sensitivity labels and label policies, the custom part/drawing-number sensitive-info type, DLP policies (with mode: test/audit vs enforce), and retention labels/policies. This is the "what exists today" read used to plan the taxonomy, the DLP audit gate, and retention scope. HOW TO RUN Install-Module ExchangeOnlineManagement -Scope CurrentUser powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Discovery.ps1 Sign in as a Compliance Administrator (or Information Protection + DLP roles) when prompted. EXAMPLE .\PURVIEW-MFG-Discovery.ps1 EXPECTED RESULTS Lists of sensitivity labels (name, priority, encryption), label policies, the custom SIT 'SIT - MFG Part/Drawing Number' (present/absent), DLP policies (name, MODE — flags any already in Enforce), and retention labels/policies. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-LabelPolicy / Get-DlpSensitiveInformationType / Get-DlpCompliancePolicy / Get-ComplianceTag / Get-RetentionCompliancePolicy. Availability is licensing-/cloud-dependent. > [CmdletBinding()] param([string]$SitName = 'SIT - MFG Part/Drawing Number')

How to run: Install-Module ExchangeOnlineManagement -Scope CurrentUser powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Discovery.ps1 Sign in as a Compliance Administrator (or Information Protection + DLP roles) when prompted. EXAMPLE .\PURVIEW-MFG-Discovery.ps1 EXPECTED RESULTS Lists of sensitivity labels (name, priority, encryption), label policies, the custom SIT 'SIT - MFG Part/Drawing Number' (present/absent), DLP policies (name, MODE — flags any already in Enforce), and retention labels/policies. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-LabelPolicy / Get-DlpSensitiveInformationType / Get-DlpCompliancePolicy / Get-ComplianceTag / Get-RetentionCompliancePolicy. Availability is licensing-/cloud-dependent. > [CmdletBinding()] param([string]$SitName = 'SIT - MFG Part/Drawing Number')

PURVIEW-MFG-Remediation.ps1

Purpose: Remediates the low-risk gaps that PURVIEW-MFG-Validation.ps1 finds — enabling a published-but- off Manufacturing label policy and enabling a disabled retention policy — WITHOUT ever silently switching a DLP policy to enforce (that is the audit gate, a human decision) and without touching label encryption or the export-control rule. Runs in -WhatIf by default; pass -Apply to make changes. It never deletes anything. HOW TO RUN Preview (safe, default): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Remediation.ps1 Apply changes: powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Remediation.ps1 -Apply Sign in as a Compliance Administrator when prompted. EXAMPLE .\PURVIEW-MFG-Remediation.ps1 -WhatIf EXPECTED RESULTS A list of proposed actions (WhatIf) or the applied changes (-Apply). DLP enforce is never automated — the script only reports DLP policies still in test/audit for the operator to promote via the Implementation Guide (Step 8). .NOTES White-label — Purview Solutions. Default is a dry run. PowerShell 5.1+/7+. Uses Get-/Set-LabelPolicy and Get-/Set-RetentionCompliancePolicy; reads Get-DlpCompliancePolicy. > [CmdletBinding()] param([switch]$Apply, [switch]$WhatIf, [string]$LabelPolicyName = 'Label Policy - Manufacturing Taxonomy - All Users', [string]$RetentionPolicyName = 'Retention Policy - Manufacturing Baseline - 5 Year') $doApply =

$Apply -and -not $WhatIf $mode = $(if($doApply){'APPLY'}else{'WHATIF (preview only)'})

How to run: Preview (safe, default): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Remediation.ps1 Apply changes: powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Remediation.ps1 -Apply Sign in as a Compliance Administrator when prompted. EXAMPLE .\PURVIEW-MFG-Remediation.ps1 -WhatIf EXPECTED RESULTS A list of proposed actions (WhatIf) or the applied changes (-Apply). DLP enforce is never automated — the script only reports DLP policies still in test/audit for the operator to promote via the Implementation Guide (Step 8). .NOTES White-label — Purview Solutions. Default is a dry run. PowerShell 5.1+/7+. Uses Get-/Set-LabelPolicy and Get-/Set-RetentionCompliancePolicy; reads Get-DlpCompliancePolicy. > [CmdletBinding()] param([switch]$Apply, [switch]$WhatIf, [string]$LabelPolicyName = 'Label Policy - Manufacturing Taxonomy - All Users', [string]$RetentionPolicyName = 'Retention Policy - Manufacturing Baseline - 5 Year') $doApply = $Apply -and -not $WhatIf $mode = $(if($doApply){'APPLY'}else{'WHATIF (preview only)'})

PURVIEW-MFG-Report.ps1

Purpose: Summarizes information-protection coverage for the Manufacturing IP + CUI program so you can review the DLP audit gate and track classification: (1) which sensitivity labels exist and their protection, (2) which services each DLP policy covers and its mode, and (3) recent DLP events from the unified audit log grouped by workload — the audit data you tune before enforcing (Implementation Step 7). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Report.ps1 -Days 7 -CsvPath C:\Temp\mfg-dlp.csv Sign in as a Compliance Administrator when prompted. Requires audit logging enabled (Requirement 2.7) for the DLP-event section to return data. EXAMPLE .\PURVIEW-MFG-Report.ps1 -Days 14 EXPECTED RESULTS A label inventory, a DLP location/mode matrix (including endpoint), and a per-workload count of DLP rule matches for the window, optionally exported to CSV. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-DlpCompliancePolicy and Search-UnifiedAuditLog (RecordType DLP). > [CmdletBinding()] param([int]$Days = 7, [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Report.ps1 -Days 7 -CsvPath C:\Temp\mfg-dlp.csv Sign in as a Compliance Administrator when prompted. Requires audit logging enabled (Requirement 2.7) for the DLP-event section to return data. EXAMPLE .\PURVIEW-MFG-Report.ps1 -Days 14 EXPECTED RESULTS A label inventory, a DLP location/mode matrix (including endpoint), and a per-workload count of DLP rule matches for the window, optionally exported to CSV. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-DlpCompliancePolicy and Search-UnifiedAuditLog (RecordType DLP). > [CmdletBinding()] param([int]$Days = 7, [string]$CsvPath)

PURVIEW-MFG-Set-ScopeTargeting.ps1

Purpose: Creates or updates the Microsoft Entra security group used to pilot the Manufacturing Purview program ('Purview - Manufacturing Pilot') and, optionally, adds a supplied list of pilot users (engineering + a CUI/contracts team). This is the pilot scope every label/DLP/retention policy targets before tenant-wide expansion. Runs in -WhatIf by default; pass -Apply to make changes. It never deletes the group or removes members. HOW TO RUN Preview (safe, default): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Set-ScopeTargeting.ps1 Apply (create/update): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Set-ScopeTargeting.ps1 -Apply Add members: .\PURVIEW-MFG-Set-ScopeTargeting.ps1 -Apply -Members user1@contoso.com,user2@contoso.com Sign in with a Groups Administrator (or better) when Connect-MgGraph prompts. EXAMPLE

.\PURVIEW-MFG-Set-ScopeTargeting.ps1 -WhatIf EXPECTED RESULTS Reports whether the group exists and what would change (WhatIf) or creates/updates it (-Apply), then reports membership. Use the group as the pilot scope in the label/DLP/retention policies. .NOTES White-label — Purview Solutions. Default is a dry run. PowerShell 5.1+/7+. Microsoft Graph: Group.ReadWrite.All (+ User.Read.All to resolve members). No credentials in code. > [CmdletBinding()] param([string] $GroupName = 'Purview - Manufacturing Pilot', [string] $MailNickname = 'purview-manufacturing-pilot', [string[]]$Members, [switch] $Apply, [switch] $WhatIf) $doApply = $Apply -and -not $WhatIf $mode = $(if($doApply){'APPLY'}else{'WHATIF (preview only)'})

How to run: Preview (safe, default): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Set-ScopeTargeting.ps1 Apply (create/update): powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Set-ScopeTargeting.ps1 -Apply Add members: .\PURVIEW-MFG-Set-ScopeTargeting.ps1 -Apply -Members user1@contoso.com,user2@contoso.com Sign in with a Groups Administrator (or better) when Connect-MgGraph prompts. EXAMPLE .\PURVIEW-MFG-Set-ScopeTargeting.ps1 -WhatIf EXPECTED RESULTS Reports whether the group exists and what would change (WhatIf) or creates/updates it (-Apply), then reports membership. Use the group as the pilot scope in the label/DLP/retention policies. .NOTES White-label — Purview Solutions. Default is a dry run. PowerShell 5.1+/7+. Microsoft Graph: Group.ReadWrite.All (+ User.Read.All to resolve members). No credentials in code. > [CmdletBinding()] param([string] $GroupName = 'Purview - Manufacturing Pilot', [string] $MailNickname = 'purview-manufacturing-pilot', [string[]]$Members, [switch] $Apply, [switch] $WhatIf) $doApply = $Apply -and -not $WhatIf $mode = $(if($doApply){'APPLY'}else{'WHATIF (preview only)'})

PURVIEW-MFG-Validation.ps1

Purpose: Verifies that the named objects this solution deploys are present and in the expected state: the Manufacturing sensitivity labels (MFG-*) and label policy are published, the custom part/drawing-number SIT exists, the DLP policy exists (and reports its mode so you can see whether it has passed the audit gate into enforce), and the retention policy exists and is enabled. Pass/fail per the Testing Guide. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Validation.ps1 Sign in as a Compliance Administrator when prompted. Optionally pass -LabelPolicyName / -DlpPolicyName / -RetentionPolicyName / -SitName to match your renamed objects. EXAMPLE .\PURVIEW-MFG-Validation.ps1 EXPECTED RESULTS "VALIDATION PASSED" when the label policy is enabled, the custom SIT is present, the DLP policy is present, and the retention policy is enabled. Exit code 0 on pass, 1 on any gap. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-LabelPolicy / Get-DlpSensitiveInformationType / Get-DlpCompliancePolicy / Get-RetentionCompliancePolicy. > [CmdletBinding()] param([string]$LabelPolicyName = 'Label Policy - Manufacturing Taxonomy - All Users', [string]$SitName = 'SIT - MFG Part/Drawing Number', [string]$DlpPolicyName = 'DLP Policy - Manufacturing IP and CUI', [string]$RetentionPolicyName = 'Retention Policy - Manufacturing Baseline - 5 Year')

How to run: powershell -ExecutionPolicy Bypass -File .\PURVIEW-MFG-Validation.ps1 Sign in as a Compliance Administrator when prompted. Optionally pass -LabelPolicyName / -DlpPolicyName / -RetentionPolicyName / -SitName to match your renamed objects. EXAMPLE .\PURVIEW-MFG-Validation.ps1 EXPECTED RESULTS "VALIDATION PASSED" when the label policy is enabled, the custom SIT is present, the DLP policy is present, and the retention policy is enabled. Exit code 0 on pass, 1 on any gap. Read-only; no changes made. .NOTES White-label — Purview Solutions. Read-only. PowerShell 5.1+/7+. Uses Get-Label / Get-LabelPolicy / Get-DlpSensitiveInformationType / Get-DlpCompliancePolicy / Get-RetentionCompliancePolicy. > [CmdletBinding()] param(

```
[string]$LabelPolicyName = 'Label Policy - Manufacturing Taxonomy - All Users', [string]$SitName = 'SIT - MFG Part/Drawing Number', [string]$DlpPolicyName = 'DLP Policy - Manufacturing IP and CUI', [string]$RetentionPolicyName = 'Retention Policy - Manufacturing Baseline - 5 Year' )
```